

QUASI STATIC PROCESS

In thermodynamics, a quasi-static process is a process that occurs infinitely slowly, so that the system remains in thermodynamic equilibrium at all time. This means the system's state variables (like pressure, volume, and temperature) are well-defined at any time and change gradually and smoothly.

Let's talk about different laws followed by ideal gas

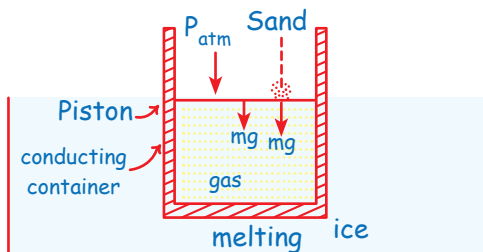
1. Boyle's Law

For a fixed amount of gas at constant temperature, pressure is inversely proportional to volume

$$P \propto \frac{1}{V} \text{ or } PV = \text{constant}$$

Setup

- ★ A gas is trapped in a conducting container with a movable piston.
- ★ The system is maintained at constant temperature (by using melting ice). It can maintain constant temperature
- ★ External pressure is applied by atmospheric pressure P_{atm} and additional weight (sand) on the piston.

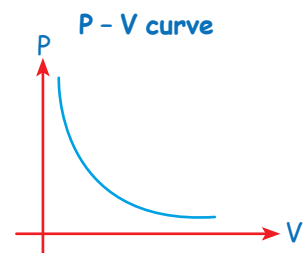


Effect of Volume Change

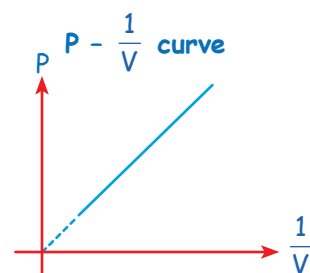
- ★ When pressure increases (by adding weight), the volume decreases as gas molecules are compressed.
- ★ When pressure decreases (by removing weight), the volume increases as gas expands.

P (Pressure) increase $\rightarrow V$ (Volume) decreases

P (Pressure) decrease $\rightarrow V$ (Volume) increases



A hyperbolic relationship, illustrating that pressure decreases as volume increases and vice versa.



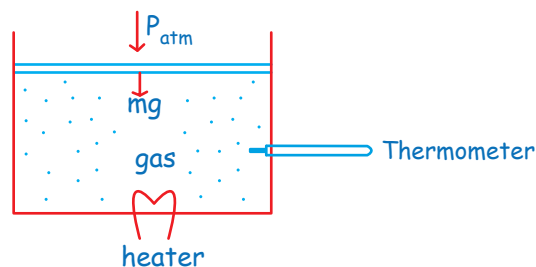
A linear relationship, indicating direct proportionality between pressure and the reciprocal of volume

2. Charles's Law

- ★ For a fixed amount of gas at constant pressure. The volume of the gas is directly proportional to its absolute temperature.

$$V \propto T$$

- ★ The system consists of a gas trapped under a movable piston, with a thermometer to measure temperature.
- ★ The gas is subjected to constant external pressure (P_{atm}), with an additional force due to the weight mg of the piston.



Effect of Heating

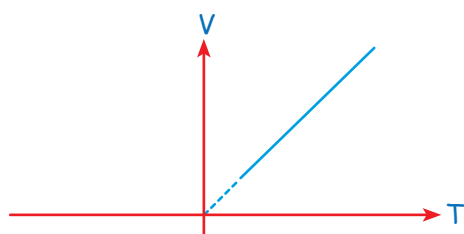
★ When heat is supplied, the gas molecules gain kinetic energy and move faster.

This increased motion causes the gas to expand, pushing the piston upward.

★ Since pressure is constant, the only way for the gas to balance the increased thermal energy is by increasing its volume.

$$\text{i.e., } P_{\text{gas}} = P_{\text{atm}} + \frac{mg}{A} \Rightarrow \text{Pressure remains unchanged}$$

V-T curve



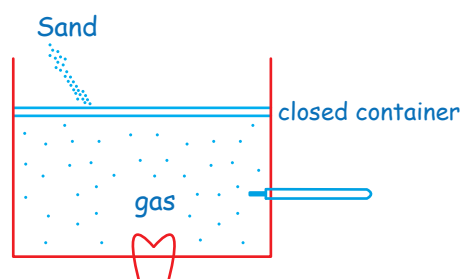
3. Gay Lussac's law

No. of moles and volume $\rightarrow$ fixed $= P \propto T$

A gas is enclosed in a fixed-volume container.

The number of gas molecules remains constant.

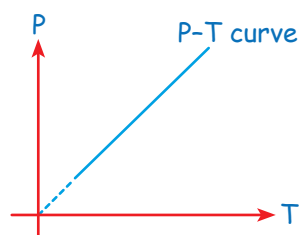
Heat is supplied using a burner, increasing the gas temperature



Effect of Temperature Change

When temperature increases, the kinetic energy of gas molecules increases.

The molecules collide more frequently and with greater force, raising the pressure.



Ideal Gas Equation

The ideal gas law describes the relationship between pressure, volume, temperature, and the number of moles of an ideal gas.

$$PV = nRT$$

Where

P = Pressure (in Pascals, Pa)

V = Volume (in cubic meters, m^3)

n = Number of moles of gas

R = Universal Gas Constant = $8.314 \text{ J}/(\text{mol}\cdot\text{K})$

T = Temperature (in Kelvin, K)

R : $8.314 \text{ J/mol} - \text{K}$

$$\frac{25}{3} \text{ J/mol} - \text{K}$$

Moles of a gas(n)

$$n = \text{no. of moles} = \frac{\text{mass of gas}}{\text{molar mass}} = \frac{m}{M}$$

$$\left[\begin{array}{l} m \text{ in gm or kg} \\ M \text{ in g/mol or kg(mol)} \end{array} \right]$$

Put in $n = \frac{m}{M}$ into ideal gas law

$$PV = \frac{m}{M} RT$$

$$PM = \rho RT \quad (\because \text{density } \rho = \frac{m}{V})$$

$$\text{Boltzmann constant } k = \frac{R}{N_A} = 1.38 \times 10^{-23} \text{ J/K}$$

$$N_A = 6.022 \times 10^{23}$$

Q. (i) In a container of 5l, we have 1.4g N_2 gas at 1800K. Find pressure of gas

(ii) If 30% gas dissociates into nitrogen atom, find P_{final} .

Sol. (i) From $P \times (5 \times 10^{-3}) \text{ m}^3$

$$= \frac{1.4}{28} \text{ mol} \times 8.314 \text{ Pa m}^3 / \text{mol} - \text{K} \cdot 1800 \text{ K}$$

$$P = 1.5 \times 10^5 \text{ Pa.}$$

(ii) Let initially there are 100 particles

$$\text{Finally, } 70 + 2 \times 30 = 130$$

$$\therefore \frac{1.5 \times 10^5}{100} = \frac{P_f}{130} \quad (\because P \propto N)$$

$$P_f = 1.9 \times 10^5 \text{ Pa}$$

Q. At some initial temperature 8g O_2 , was at pressure P . Now, it is heated by 50K and to maintain same pressure, 1.6g O_2 is released. What was the Initial temperature?

Sol. $PV = \frac{8}{32} RT$

$$PV = \frac{6.4}{32} R(T+50)$$

Solving, $T = 200 \text{ K}$

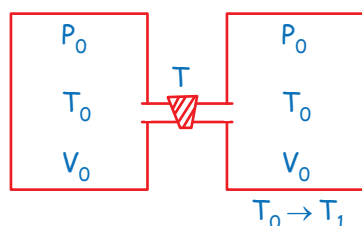
Q. An ideal gas follows $PV^2 = \text{const.}$ what will happen to its temp. when volume is doubled.

Sol. $(PV)V = \text{const}$

$$nRTV = \text{const} \Rightarrow T \propto \frac{m}{M}$$

$$V = 2V \Rightarrow T = \frac{T}{2} \quad \text{Temp} \rightarrow \text{half}$$

Q. A container is divided into two equal compartments by a valve. Each side initially has a gas at pressure P_0 , volume V_0 , and temperature T_0 . The valve is then opened, and the temperature on one side changes to T_1 . What will be the final pressure P in both compartments?



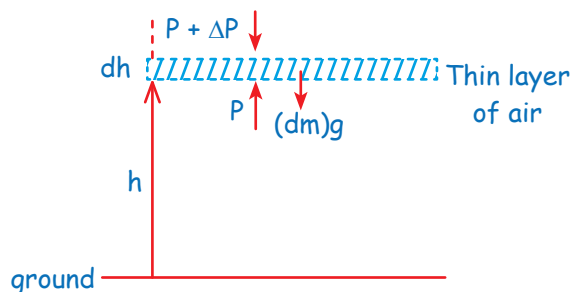
Sol. $n_i = n_f$

$$2 \times \frac{P_0 V_0}{RT_0} = \frac{PV_0}{RT_0} + \frac{PV_0}{RT_1}$$

$$P = \frac{2P_0 T_1}{T_0 + T_1}$$

Q. How does atmospheric pressure vary with height assuming constant temperature?

Sol. Consider a thin layer of air as shown in figure at height h above ground mass of this layer is dm



Balancing forces on this layer we get

$$(P + dP)A + (dm)g = PA$$

$$dP = \frac{-(dm)g}{A} = \frac{-\rho(d\text{vol})g}{A}$$

$$dP = -\rho \cdot (dh)g$$

$$PM = \rho RT$$

$$\Rightarrow dP = -\frac{PM}{RT}(dh)g$$

$$\int_{P_0}^P \frac{dP}{P} = \frac{-gM}{RT} \int_0^h dh$$

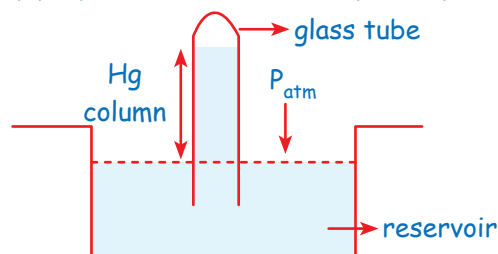
$$P = P_0 \cdot e^{\frac{-Mgh}{RT}}$$

Barometric formula

BAROMETER

A barometer is an instrument used to measure atmospheric pressure. A mercury barometer consists of a glass tube sealed at one end and filled with mercury, which is then inverted into a reservoir of mercury.

The height of the mercury column in the tube is directly proportional to the atmospheric pressure.



At equilibrium

$$\frac{mg}{A} = P_{\text{atm}} \quad (m \text{ is mass of mercury column, } A \text{ is cross-sectional area of tube})$$

$$P_{\text{atm}} = \frac{\rho V g}{A} = \frac{\rho A h g}{A} \quad (\because mg = \rho V g \text{ and } V = Ah)$$

$$\Rightarrow P_{\text{atm}} = \rho g h$$

$$h \approx 76 \text{ cm} \quad \text{Atmospheric pressure at sea level}$$

$$h \approx 75.2 \text{ cm}$$

$$g \approx 9.81 \text{ m/s}^2$$

$$\rho \approx 13.6 \times 10^3 \text{ kg/m}^3 - \text{Density of Hg}$$

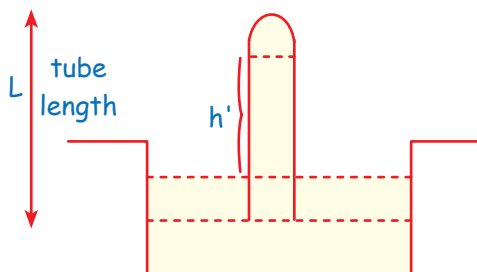
$$P_{\text{atm}} = 1.01 \times 10^5 \text{ Pa} = 1 \text{ atm}$$

FAULTY BAROMETER

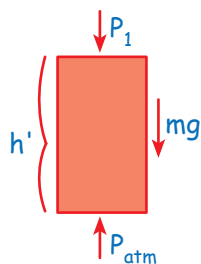
In an ideal barometer, the space above the mercury is a vacuum (zero pressure), ensuring that the height of the mercury column is purely determined by atmospheric pressure.

In a faulty barometer, a small amount of air remains trapped above the mercury column. This trapped air exerts additional pressure, altering the measurement.

Pressure measured by faulty barometer = $\rho gh'$
reading = h' cm of Hg.



reading = h' cm of Hg.



Due to the presence of trapped air, the actual pressure in side tube:

$$P_1 + \frac{mg}{A} = P_{\text{atm}}$$

$$P_1 + \rho gh' = \rho gh$$

↑
reading of correct
barometer/true reading

$$P_1 = \rho g(h - h')$$

$$\text{volume of air trapped} = V_1 = A(L - h')$$

From ideal gas law,

$$P_1 V_1 = nRT$$

$$\rho g(h - h') A(L - h') = nRT$$

Q. In a correct barometer, we inject some air of 2 cm^3 at 1 atm pressure. This air expands to 30 cm^3 inside cube. If 1 atm = 75 cm of Hg. Find reading of this faulty barometer. Cross-sectional area is 1 cm^2 .

Sol. $P_0 V_0 = P_1 V_1$

$$P_1 = \frac{1}{15} \text{ atm}$$

FBD of Hg column

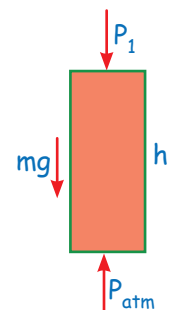
$$\frac{1}{15} + \frac{mg}{A} = P_{\text{atm}}$$

$$\frac{1}{15} + \rho gh' = 1$$

$$\rho gh' = \frac{14}{15} \text{ atm}$$

$$h' \text{ cm of Hg} = \frac{14}{15} \times 75 \text{ cm of Hg}$$

$$h' = 70 \text{ cm} \text{ Ans}$$



Q. A faulty barometer reads 68 cm and 70 cm when true values are 72 cm and 75 cm. Find total length of tube.

Sol. For faulty barometer,

$$\rho gA(h - h')(L - h') = nRT$$

$$(h - h')(L - h') = \frac{nRT}{\rho gA} = \text{const.}$$

$$(72 - 68)(L - 68) = (75 - 70)(L - 70)$$

$$L = 78 \text{ cm} \text{ Ans}$$

KINETIC THEORY OF GASES (KTG)

Assumptions

- ★ Vol. of gas = vol. of container
- ★ A gas consists of large no. of identical tiny, elastic particles.
- ★ Gas particles move randomly with all possible velocities in all possible directions.
- ★ Size of particles is much smaller than the avg. separation betⁿ them.
- ★ Gas particles don't exert any force from a distance. But they exert force only during collision.
- ★ All collisions are elastic
- ★ They follow NLM
- ★ Collision time is very small as compared to time, interval between collisions.

PRESSURE IN GAS

- ★ The pressure exerted by an ideal gas arises from the continuous collisions of gas molecules with the walls of the container.
- ★ the pressure exerted by an ideal gas is directly proportional to the average kinetic energy of its molecules and is given by the formula

$$P = (1/3) * (\rho * V_{rms}^2),$$

where P is pressure, ρ is density, and v^2 is the mean square speed.

$P = 1/3 * (m_{gas}/V) * V_{rms}^2$ Where V is volume of container and m_{gas} is mass of gas inside container

Force on wall = pressure * area

RMS' VELOCITY

$$\sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m}}$$

Translational Kinetic Energy

- ★ The average translational kinetic energy (KE) of a gas molecule is directly proportional to the absolute temperature (T) and is given by the formula

$$KE = (3/2)kT,$$

where 'k' is the Boltzmann constant.

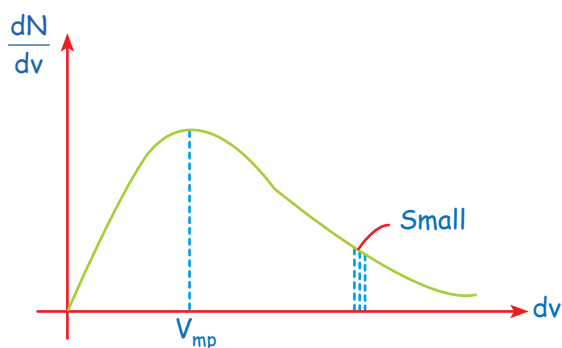
T is temperature in kelvin

Avg. translational KE for 1 molecule =

$$\frac{1}{2} m V_{rms}^2 = \frac{3}{2} kT$$

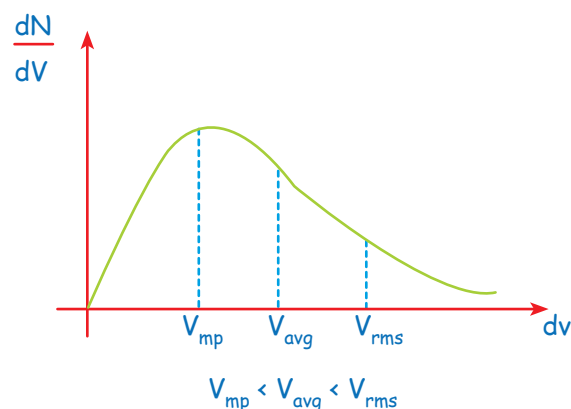
- ★ Avg. translational KE for 1 mole = $\frac{1}{2} m V_{rms}^2 = \frac{3}{2} RT$

MAXWELL'S SPEED DISTRIBUTION CURVE



area = $\frac{dN}{dv} dv = dN$ = no. of particles moving with velocity v to $v + dv$

Total area of graph = $\int dN = N$ = total no. of particles



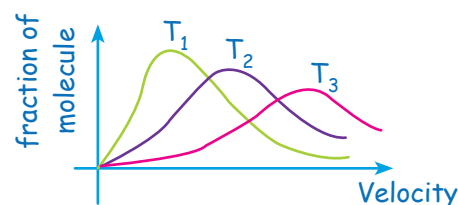
Average speed: the mean speed of all molecule

$$\text{i.e., } V_{avg} = \frac{\int v \cdot (dN)}{\int dN} = \sqrt{\frac{8 RT}{\pi M}} = \sqrt{\frac{8 kT}{\pi m}}$$

Most Probable Speed (V_{mp}): The speed at which the highest number of molecules are moving.

$$V_{mp} = \sqrt{\frac{2RT}{M}} \quad (\text{Most Probable Speed})$$

Q.



Sol. $T_3 > T_2 > T_1$

Q. Given the following velocities of gas molecules:
 $v = 0, 2, 3, 4, 4, 4, 5, 5, 6, 9$ m/s, calculate the following:

1. Average velocity (V_{avg})
2. Root mean square velocity (V_{rms})
3. Most probable velocity (V_{mp})

Sol. $V_{avg} = \frac{42}{10} = 4.2$ m/s

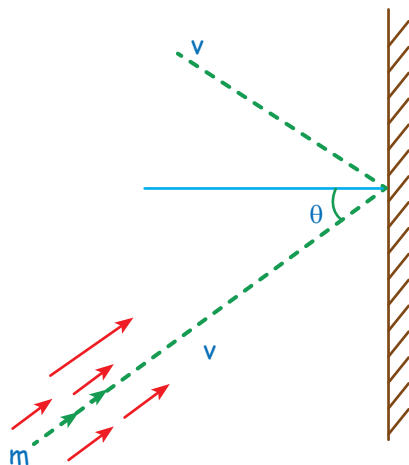
$$V_{rms} = \sqrt{\frac{0+4+9+48+50+36+81}{10}} = \sqrt{22.8} \text{ m/s.}$$

$$V_{mp} = 4 \text{ m/s.}$$

Q. We have a parallel beam of gas particles each mass m and speed v is incident on a wall at an angle θ with respect normal direction.

The particles density is n per unit volume. Find the pressure on wall. [Assume elastic collision]

Sol. $\Delta P_{1 \text{ particle}} = 2mv \cos \theta$



For N particles $= 2(mN)v \cos \theta$

$$F = \frac{N2mv \cos \theta}{\Delta t}$$

N is no. of particles striking in a time int Δt

$$N = n(A \cdot l) \quad n = \frac{N}{\text{volume}}$$

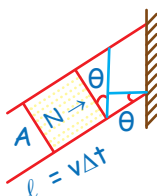
$$N = nAv \Delta t$$

$$F = \frac{nAv \Delta t \cdot 2mv \cos \theta}{\Delta t}$$

$$= 2mnv^2 (\cos \theta) A$$

$$P = \frac{F}{\text{Area}} = \frac{F}{A_{\text{wall}}} = \frac{F}{A / \cos \theta}$$

$$P = 2mnv^2 \cos^2 \theta$$



Special Case Normal Incidence

$$\theta = 0$$

$$P = 2mnV^2$$

If wall surface is also moving, $P = 2mn(v_{\text{rel}})^2$

↓
vel. of particles w.r.t. wall.

MEAN FREE PATH

A moving particle sweeps out cylindrical Effective interaction volume in time $\Delta t = (\pi d^2) \cdot V(\Delta t)$

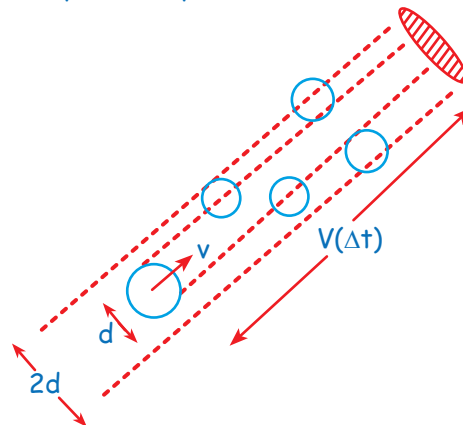
Where,

d = diameter of a particle (collision cross-section),

V = molecular velocity,

Δt = small time interval.

$n_v \rightarrow$ no. of particles per unit vol.



No. of particles in a volume $= n_v \times \text{volume} = n_v(\pi d^2)$
 $V(\Delta t) = N$

$\therefore$ No of collision $= N$

Avg. time interval between successive collision

$$\tau = \frac{\Delta t}{N} = \frac{\Delta t}{n_v \cdot (\pi d^2) V \Delta t} = \frac{1}{n_v \cdot \pi d^2 V}$$

↓
 V is avg. velocity

$$\tau_{\text{avg}} = \frac{1}{n_v (\pi d^2) V_{\text{avg}}}$$

↓
relaxation time

Mean free path: Avg. dist. travelled between two successive collisions

$$\lambda_m = V_{\text{avg}} \cdot \tau_{\text{avg}}$$

$$= \frac{1}{n_v \pi d^2}$$

n_v = no. of particles per unit volume

$$\therefore n_v = \frac{N}{\text{vol}} = \frac{n N_A}{\text{vol.}}$$

↓
no. of moles

↑
Avogadro's number

Therefore, mean free path

$$\lambda_m = \frac{1}{\left(\frac{nN_A}{V}\right)\pi d^2} = \frac{\overset{\text{volume}}{V}}{nN_A\pi d^2}$$

Using $PV = nRT$

$$\lambda_m = \frac{V}{nN_A\pi d^2} = \frac{RT}{PN_A\pi d^2} = \frac{kT}{P\pi d^2} \quad \left(\because k = \frac{R}{N_A}\right)$$

This is valid when all other particles are at rest. But if they are also moving, then then average relative velocity need to be taken.

$$\therefore \tau_{avg} = \frac{1}{n_v\pi d^2(v_{rel})_{avg}}$$

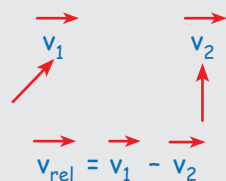
$$\& (V_{rel})_{avg} = \sqrt{2} V_{avg}$$

$$\tau_{avg} = \frac{1}{n_v\pi d^2(v_{avg})\sqrt{2}}$$

$$\lambda_m = \frac{1}{n_v(\pi d^2)\sqrt{2}}$$

$$\lambda_m = \frac{V}{nN_A(\pi d^2)\sqrt{2}} = \frac{RT}{PN_A\pi d^2\sqrt{2}} = \frac{kT}{P\pi d^2\sqrt{2}}$$

Proof



$$|v_r| = \sqrt{v_1^2 + v_2^2 - 2v_1v_2 \cos \theta}$$

$$|v_{rel}|_{avg} = \sqrt{v_{1\ avg}^2 + v_{2\ avg}^2 - 2v_{1\ avg} v_{2\ avg} \cos \theta_{avg}}$$

$$(v_1^2)_{avg} = (v_2^2)_{avg} \quad [\text{Random for all particles}]$$

$$= v_{avg}^2$$

$$(\cos \theta)_{avg} = \frac{\sum_{\theta=0^\circ}^{\theta=180^\circ} \cos \theta}{n} = 0$$

$$|v_{rel}|_{avg} = \sqrt{v_{avg}^2 + v_{avg}^2 + 0}$$

$$|v_{rel}|_{avg} = \sqrt{2} v_{avg}$$

ENERGY OF IDEAL GAS

Internal energy of gas is only KE.

And this KE consists of

$$(1) \text{ Translational KE} = \frac{3}{2}PV = \frac{3}{2}nRT$$

(2) Rotational KE

(3) Vibrational KE

Maxwell's Law of Equipartition of Energy

It says that, in thermal eq^m, the total KE of a gas molecule is equally divided among all degree of freedom & energy associated with each degree of

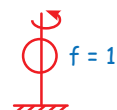
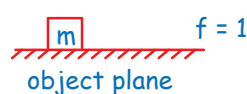
freedom is $= \frac{1}{2}kT$

For a system of f no. of degree of freedom, avg. energy of one particle $= \frac{f}{2}kT$

Degree of Freedom

No. of independent ways in which a molecule or object can exhibit motion or store energy.

Object on a straight line



Rolling motion with slipping

$$\frac{v}{R} \neq \omega \quad \begin{matrix} 1 \text{ translational} \\ + \\ 1 \text{ Rotational} \end{matrix}$$

If pure, rolling $v = \omega R$ & $a = \alpha R$ then, $f = 1$

For Translational KE

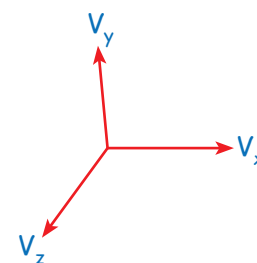
A free particle moving in three-dimensional space has three translational degrees of freedom (along x , y , and z). Kinetic energy for translation

$$KE_{tr} = \frac{1}{2}mv^2$$

Expanding in three dimension

$$K.E_{tr} = \frac{1}{2}mV_x^2 + \frac{1}{2}mV_y^2 + \frac{1}{2}mV_z^2$$

$f = 3$



For Rotational KE,

The rotational degrees of freedom depend on the structure and constraints of the object:

A linear molecule has $f = 2$ (rotation about two perpendicular axes).

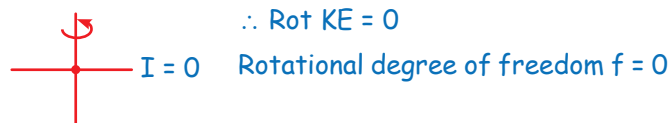
A non-linear molecule has $f = 3$ (rotation about all three perpendicular axes).

Rotational kinetic energy:

$$KE_{\text{rot}} = \frac{1}{2} I \omega^2$$

Monoatomic Gases

A monoatomic gas (e.g., noble gases like He, Ne, Ar) has only translational degrees of freedom.



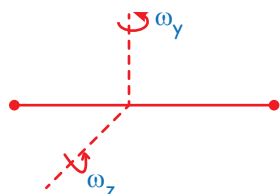
$\therefore$ Rot KE = 0

Rotational degree of freedom $f = 0$

Translational degree of $f = 3$

Diatomic Gases

A diatomic molecule (e.g., O_2 , N_2 , HCl) consists two atoms connected by a bond.



In addition to three translational degrees of freedom, a diatomic molecule has two additional rotational degrees of freedom about axes perpendicular to the molecular axis.

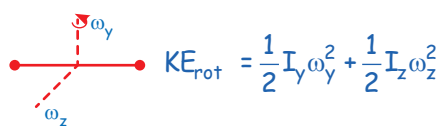
The rotational kinetic energy:

$$KE_{\text{rot}} = \frac{1}{2} I_y \omega_y^2 + \frac{1}{2} I_z \omega_z^2$$

$f = 2$ (rotational) + 3 (translational)

Triatomic Gases. (CO_2 , H_2O)

(a) linear triatomic



$f = 2$ (rotational) + 3 (translational)

(b) non-linear triatomic



$f = 3$ (rotational) + 3 (translational)

Vibrational Motion: Valid for Very High Temperature



Energy associated with vibrational motion.

$$E = \frac{1}{2} kx^2 + \frac{1}{2} \mu v_{\text{rel}}^2$$

Energy stored in vibration bond

Energy due to molecular vibration

Degree of freedom $f = 2$

★ Each vibrational mode is associated with 2 degree of freedom.

★ No of vibrational modes = $(3N - 5)$ for linear molecule

= $(3N - 6)$ for a non - linear molecule

Where, N = no. of atoms

★ For diatomic mol., $N = 2$

No of vibrational modes = $3(2) - 5 = 1$

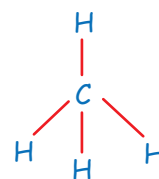
Vibrational degree of freedom = 2

Total no. of degree of freedom = $\frac{3}{\text{tr}} + \frac{2}{\text{rot}} + \frac{2}{\text{vib}} = 7$



Q. Find total no. of degree of freedom. For CH_4

Sol. $f_{\text{tr}} = 3$, $f_{\text{rot}} = 3$



modes of vib = $3(5) - 6 = 9$

$\therefore f_{\text{vib}} = 9 \times 2 = 18$

Total $f = 24$ Ans.

Additional explanation.

According to equipartition law, each degree of freedom has $\frac{1}{2} kT$ energy. So, internal energy for a gas for N particles

$$= \left(\frac{f}{2} kt \right) N$$

Note: (U is KE only)

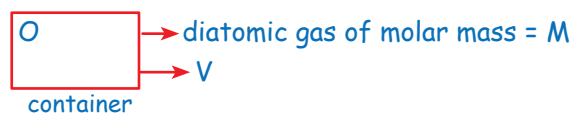
$$= \frac{f}{2} kt(nN_A) = \frac{f}{2} nRT = \frac{f}{2} PV$$

Q. Find the approx energy of air in our room. Size of room is $3\text{m} \times 5\text{m} \times 5\text{m}$

Sol. $f = 5$. (For air-mainly consists O_2 & N_2)

$$E = \frac{f}{2} PV = \frac{5}{2} \times 10^5 \times 75 = 1.875 \times 10^7 \text{ J}$$

Q. If the container of a diatomic gas (with molar mass M) suddenly stops, find the change in the temperature of the gas.



Sol. Loss in energy of moving gas is converted into internal energy leading to change in temperature

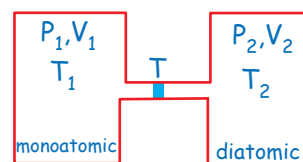
$$\frac{1}{2} (m_{\text{gas}}) v^2 = \Delta U = \frac{f}{2} nR(\Delta T)$$

$$\frac{1}{2} mNv^2 = \frac{5}{2} nR(\Delta T)$$

$$\frac{1}{2} mnN_A v^2 = \frac{5}{2} nR(\Delta T)$$

$$\Delta T = \frac{MV^2}{5R}$$

Q. If both containers are insulated from surrounding & value is removed. Find final Pressure & Temperature on both sides.



Sol. Moles conserved:

$$n_1 = n_2$$

$$\frac{P_1 V_1}{T_1} + \frac{P_2 V_2}{T_2} = \frac{P}{T} (V_1 + V_2)$$

$\therefore$ Insulated containers, $U_i = U_f$ (energy conserved)

$$\frac{3}{2} n_1 R T_1 + \frac{5}{2} n_2 R T_2 = \left(\frac{3}{2} n_1 R + \frac{5}{2} n_2 R \right) T$$

$$T = \frac{3n_1 T_1 + 5n_2 T_2}{3n_1 + 5n_2} = \frac{3P_1 V_1 + 5P_2 V_2}{\frac{3P_1 V_1}{T_1} + \frac{5P_2 V_2}{T_2}}$$

Degree of freedom for mix. of gases.

$$U_{\text{mix}} = U_1 + U_2$$

$$\left(\frac{f_{\text{mix}}}{2} \right) n_{\text{mix}} RT = \frac{f_1 n_1 RT}{2} + \frac{f_2 n_2 RT}{2}$$

$$f_{\text{mix}} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}$$

$$f_{\text{mix}} = \frac{\sum n_i f_i}{\sum n_i}$$